New Green Synthesis and Formulations of Acyclovir Prodrugs

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Different green synthesis of alkyl esters of acyclovir (acyclovir prodrugs) is described. Hexanoic, decanoic, dodecanoic and tetradecanoic acyclovir esters were synthesized reacting acyclovir and the respective acid anhydride in dimethyl sulfoxide (DMSO), in solvents from renewable sources and without solvent (T=30 °C). Yields in prodrugs after 10 min of reaction were >95% using DMSO as solvent. The purification methodology was very simple, shorter and greener than previously described. The biosolvent, *N*,*N*-dimethylamide of decanoic acid, let us to obtain >95% yield at 24 h. This oily biosolvent is not dermotoxic and the reaction crude can directly be used in topic formulations. Syntheses without solvent proceeded successfully for acyclovir esters. Indeed, dodecanoate and tetradecanoate yielding >98% conversion of reactants in 30 min. In spite of requiring mild temperature (65 °C), substrate molar ratios were lowered to 1 : 1, thus conducing to a more efficient use of raw materials. The synthetic procedures were scaled up to a 300 g batch (yield 98—99% isolated ester). These esters can be used as acyclovir prodrugs in topic formulations. The esters release from an oil/water micro-emulsion and a hydrogel formulation were tested with good results.

Key words acyclovir prodrug; green chemistry; transdermal formulation

Acyclovir is a potent inhibitor of Herpes Simplex which also shows relatively fair inhibition to Varizella Zoster virus and to a less extent to Epstein Barr virus[1] Its wide and continued clinical use as antiherpetic agent during the last 25 years has proved its safety[2] Current antiherpetic treatments with acyclovir include tablets (200 mg, 400 mg, 800 mg or 1 g), topical cream (5%), intravenous injection (25 mg/ml) and ophthalmic ointment (3%). Nevertheless, acyclovir shows some limitations due to its low bioavailability (15—20%) and low serum half-life (2—3 h),[1,3,4] thus high doses and frequent administrations are required. Therefore, different prodrugs and topical formulations have been described in order to overcome these inconveniences.

Transdermal administration of prodrugs is an interesting path to convey active principles systemically. This methodology directly competes with oral administration. Indeed, around 40% of drugs in clinical trials are intended to be used transdermally.[5] This is due to the advantages that transdermal administration offers to achieve efficient and safe healing such as: avoiding first pass hepato-metabolic degradation,[6] giving stable drug levels in plasma,[7] this procedure is invasiveness[8] and it can be applied for longer periods of time.[9] The simplest transdermal technique is a passive diffusion which has the additional advantage of being more patient compliant than its counterparts such as iontophoretic, electroporation *etc.*, which require more complex manufacturing and administering procedures.

The most suitable kind of prodrugs for passive transdermal administration path is the linear alkyl esters due to their compatibility with the skin lipidic matrix. In addition the high activity of esterases in human skin favors the liberation of the drug by simple hydrolysis of acyclovir and the non-toxic fatty acid.

Early attempts to synthesize acyclovir prodrugs date from early 80s with the work on 2,6-diamine-9-(2-hydroxyetoximethyl)purine[10] and 6-deoxyacyclovir.[11] In 1990, acetate and hexanoate of acyclovir were synthesized in 2—3 h by Hardnen *et al.*[12] using acid anhydrides as acyl donors in *N*,*N*-dimethylformamide (DMF) (toxic for the skin) and 4-*N*,*N*-dimethylaminopyridine (DMAP) as catalyst, obtaining ≈40% yield in monoester and diacyl derivatives as secondary products. In 1994, Shao *et al.*[13] followed this methodology to synthesize some short chain esters of acyclovir but extended the reaction time up to 2—3 d without reporting yields. A similar approach was described by Bando *et al.*[14] in 1996 using acid chlorides instead of acid anhydrides as acyl donors but reduced the acyl chloride : acyclovir molar ratio to 3 : 1. This group obtained high purity compounds after 2 d and further precipitation in a NaOH solution. In a similar effort Kim *et al.*[15] in 1998 synthesized short chain

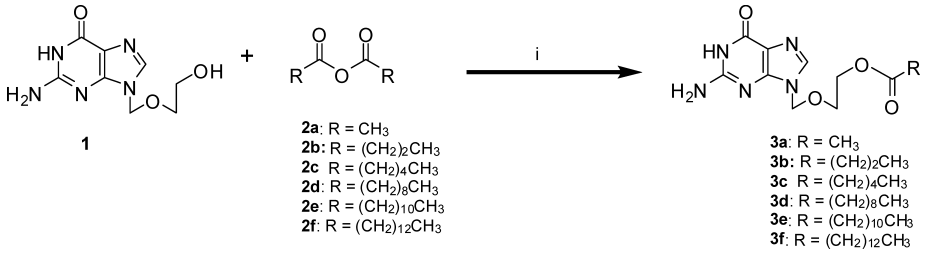


Chart 1

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esters of 2-amino-6-fluoro-9-(2-hydroxymethoxy-methyl)purine, as potential prodrugs of acyclovir. These workers described a three-step procedure with yields above 90%, but 5 d were necessary to isolate the ester. In this procedure, the esterification step was achieved in 20 min using 3 eq of acid anhydrides and DMF as solvent. In 1999, Gao *et al.*[16] conducted the synthesis of several short ester prodrugs of acyclovir using a modified Shao's procedure,[13] reducing acid anhydride amount from 20 to 3 acyclovir equivalents and the reaction time to 1.5 d. In such work they reported yields around 80%, but they still observed 5—12% of double acylations. To the best of our knowledge, the syntheses of long chain esters of acyclovir (laurate and palmitate) have been referenced only in the Gao *et al.*[17] review in which yields of 37% are reported. Recently, Stankova *et al.*[18,19] described the synthesis of acyclovir esters with peptidomimetics or with cinnamic acid using coupling reagents and evaluated *in vitro* for their antiviral activity against the replication of Herpes simplex virus type 1 (HSV-1) and type 2 (HSV-2).

Solvents are necessary substances in all the chemical organic synthesis, so attention should be paid to choose the best suited and less harmful to the environment. Therefore, organic processes which make less use or substitute toxic solvents are desired. In this way, the enzyme-catalyzed acylation of nucleosides in 2-methyltetrahydrofuarne (obtained from biomass) instead of tetrahydrofuran (THF) has been reported.[20]

In this work we present a series of alternative procedures to synthesize short and long acyclovir fatty acid esters (**3a**—**f**) in a more efficient and sustainable way to be incorporated in topical emulsions for transdermal administration use.

Results and Discussion

Synthesis of Acyclovir Esters in DMSO With the aim to carry out greener synthesis, we substituted dermotoxic DMF, used in all the described procedures above, by DMSO with low dermotoxicity and widely accepted as percutaneous penetrant and drug solvent and so, useful in creams for topical use.[21—23] The synthesis of acyclovir esteres took place at room temperature using DMSO as the solvent, 4-*N*,*N*-dimethylamino pyridine (DMAP) in catalytic amounts and acid anhydrides as the acylating agents, obtaining near quantitative yields by HPLC. Free fatty acids (with low reactivity) or acid chlorides (liberating HCl in the acylation whic decomposes the prodrugs **3a**—**f**) were not used as the acylating agents due to the low reactivity or to the dermotoxic HCl formation, respectively. The reaction time was finished in 10 min and no acylations on the amino group were observed in any case as can be deduced from NMR spectra and elemental analysis of the isolated compounds **3a**—**f**. The described purification procedure was simple and straightforward by precipitation in water. The overall time to isolate pure acyclovir ester prodrugs in a solid form was reduced from over 1.5—5 d as reported to one hour maximum improving the described syntheses. HPLC-MS analysis showed that only 0.02% of DMSO remains in the dry powder. The scope of the synthesis is general because the yields are sustained regardless of the fatty acid chain size (Table 1). This methodology outperforms the best results described by Gao and Mitra.[16,17]

Table 1. Yields Obtained in the Esterification of Acyclovir (**1**) Using DMSO as Solvent Compared with the Best Described Yields[16]

Fatty acid anhydride	Compound	Yield in isolated ester (%)	Reaction time (min)	Yield[16] (%)	Reaction time[16] (d)
Acetic	**3a**	95	10	84	1.5
Butyric	**3b**	96	10	—	—
Hexanoic	**3c**	97	10	37	1.5
Decanoic	**3d**	98	10	—	—
Dodecanoic	**3e**	97	10	37	1.5
Tetradecanoic	**3f**	98	10	—	—

Table 2. Yields Obtained after 24 h in the Synthesis of Acyclovir Alkyl Esters Using *N*,*N*-Dimethylamides of Fatty Acids as Green Solvents

	Compound	Yield (%)		
		Solvent DMAC10	Solvent DMAA	Solvent DMALA
ACVC6	**3c**	97	30	10
ACVC10	**3d**	97	NI	12
ACVC12	**3e**	98	NI	11
ACVC14	**3f**	99	NI	NI

NI: No isolated.

Synthesis of Acyclovir Esters in Solvents Obtained from Biomass An alternate approach was to use solvents obtained from renewable sources, as the reaction medium. *N*,*N*-dimethylamides of fatty acids, obtained from renewable sources (triglycerides) were used to carry out the synthesis of acyclovir esters. Three different solvents were used: *N*,*N*-dimethylamide of decanoic acid (DMAC10), a mixture of *N*,*N*-dimethylamides of octanoic and decanoic acids (DMAA) and *N*,*N*-dimethylamide of lactic acid (DMALA).

As we show in the Table 2, the three solvents provided a qualitatively positive outcome for the synthesis acyclovir alkyl esters but at slower reaction rates than in DMSO, probably due to the solvent viscosity which might create diffusion problems to substrates.[24] TLC monitoring and HPLC analysis indicated that only the monoester was synthesized. Despite this approach rendered slower reaction rates compared to DMSO, solvent DMAC10 offered a double function: as solvent and as a drug permeation enhancer for topical applications as will be discussed.

Optimization of the Synthesis Conditions In the synthesis of **3c** (used as the model), we made slight further modifications in the reaction conditions in order to improve the process. We reduced both, the substrate molar ratio acyclovir/anhydride from 1/3 to 1/2, and the amount of solvent used (DMAC10). We increased the acyclovir concentration from 250 to 875 mM. The reaction was complete at 24 h yielding almost 100% acyclovir monoester, with no signs of any sort of diacylation, identified and quantified by TLC and HPLC.

From downstream point of view, solvent and particularly catalyst, were removed pouring and agitating 30 reaction volumes of a 1/1 acetonotrile/water mixture for 5 min into the reaction medium, filtering and letting dry at room temperature. This procedure guarantees the removal of catalyst, leaving any further purification just as a means to adjust the amount of solvent remaining in the final product. From an industrial standpoint, the use of a green solvent such as

DMAC10 which serves also as a permeation enhancer for percutaneous applications, offers the advantage of avoiding the complete removal of the solvent from the reaction mixture, and use this mixture as a permeation-enhancer-included product ready to use "as is," in the following stages of transdermic formulation. In an industrial setting, the recovery and recycling of DMAC10 can be achieved much easier than the recovery of DMSO from water, due to its water immiscibility.

Following this positive outcome, the synthesis of the fatty acid acyclovir esters (**3d**—**f**) were carried out to explore the scope. In these cases, the solvent amount used was 20 ml and the temperature was raised to 40 °C to help maintain the reaction medium fluid. Reaction was completed (24 h) with yields <98% (isolated product) and only the three monoesters were obtained. The purification steps followed were the same as in the synthesis of **3c** stated above.

Synthesis without Solvent With the aim to develop a greener process, we performed the synthesis of acyclovir esters without solvent, using the anhydride as the reaction medium and using an anhydride/acyclovir ratio of 1/1. This substrate ratio was only achieved for long (C12 and C14) acyclovir anhydrides, because of their higher specific volume which, unlike shorter anhydrides, occupy a larger volume per weight and thus, are able to provide a large enough medium volume to carry out the reaction. Conversion of acyclovir into the respective ester, which reached over 98% in all cases, followed by TLC and HPLC, was achieved after 30 min of the addition of acyclovir into a melted anhydride and catalyst mix (55 °C). The monoesters of acyclovir were purified by the same methodology described above.

Scale-Up of the Synthesis Due to **3c** was the most interesting prodrug as we will show, the scale up to synthesize 300 g of **3c** was carried out dissolving 210 g (933 mmol) of 1 in 2.1 l of anhydrous DMSO with 11.3 g (93 mmol) of DMAP. After an inert atmosphere was established, 645 ml (2.79 mol) of hexanoic anhydride was added to the reaction and set vigorous agitation. After 10 min, agitation was stopped and reaction medium was poured into a suitable burette. The reaction medium was let to fall drop wise into a 4 l vessel with water in full vigorous agitation. The white precipitate supernatant was then filtered and let dry at room temperature. This precipitate purification step was repeated until consuming all reaction volume. The characterization of the reaction products was accomplished by NMR. The reaction yield was >98% (isolated product).

Prodrug Release Assays The acyclovir esters were separately incorporated into two different topical formulations: an oil/water (O/W) emulsion and a hydrogel. The release assays were performed in a Microette Plus Hanson Research using polysulfone synthetic membranes (Tuffryn®). The assay consists in measuring the amount of acyclovir ester that goes through the membrane. The acyclovir ester is embedded in a topical formulation placed in a donor chamber over the membrane. A suitable solution fills the receptor chamber under the membrane, from which samples are withdrawn.

The O/W emulsion formulation was previously described by Rodríguez-Bayón and Trujillo.[25] The percentage of each acyclovir ester was 5%. Each ester was assayed separately and the release profile obtained is shown in Fig. 1. As this figure shows, ACVC6 and ACVC12 were the acyclovir esters releasing the highest amount of acyclovir to the receptor phase (268, 197 μg, respectively).

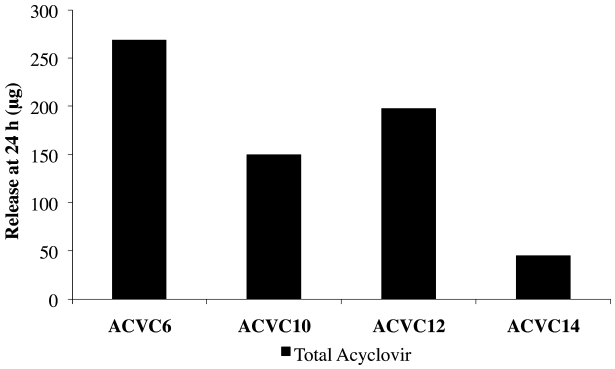


Fig. 1. Acyclovir Esters Release Assays with O/W Emulsion Using Polysulfone Synthetic Membranes (Tuffryn®)

Each ester was incorporated at 5% (w/w) into the O/W emulsion. The amount of ester that crossed the membrane was measured at 24 h.

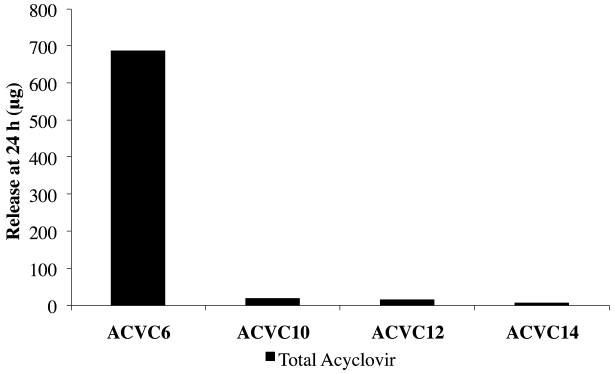


Fig. 2. Acyclovir Esters Release Assays with Hydrogel Formulation Using Polysulfone Synthetic Membranes (Tuffryn®)

Each ester was incorporated at 5% (w/w) into the hydrogel and the amounts of acyclovir in ester form that crossed the membrane into the receptor phase were measured at 24 h.

A hydrogel formulation developed by Rodríguez-Bayón and Trujillo-Casares[26] was also tested as the prodrug vehicle. The data obtained from the prodrug release assays using the hydrogel are shown in Fig. 2. ACVC6 was the only acyclovir ester able to significantly cross the membrane. In this case, the hydrogel formulation is superior to the emulsion as it enables a more than two-fold of total acyclovir release (612 μg) than W/O emulsion did.

The Green Biosolvent DMAC10 as Release Promoter As we mentioned earlier, DMAC10 is considered relatively safe solvent, according to the Environmental Protection Agency,[27] and thus, as an oily substance, and compatible with the intercellular lipidic matrix of the skin, it was tested to assess its capabilities as a permeation enhancer. The O/W already used was adapted to include ACVC6 (5% w/w) and solvent DMAC10 (5% w/w). Our results, shown in Fig. 3, indicate that solvent DMAC10 does have activity as permeation enhancer, although its enhancement factor is around 1.1 (298 μg, Fig. 3 *vs.* 268 μg, Fig. 1). This amount of acyclovir that went through the membrane is comparable to the magnitudes obtained by Bando *et al.*[14] using geranylaz-cycloheptane-2-one (GACH) as permeation enhancer in rat skin.

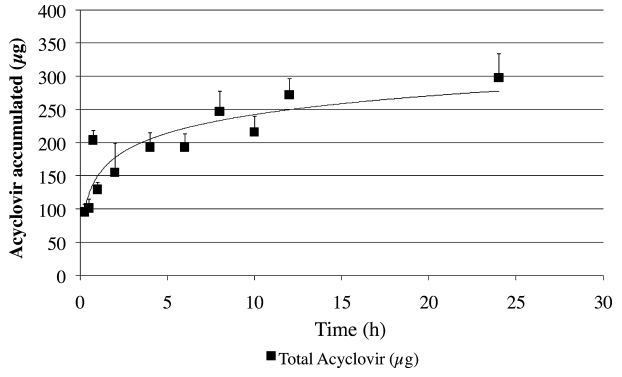


Fig. 3. Acyclovir Hexanoate Release Profile in O/W Emulsion with Solvent DMAC10 as Promotion Enhancer, Using Polysulfone Synthetic Membranes (Tuffryn®)

Formulation includes DMAC10 (5% w/w) and acyclovir hexanoate (5% w/w) into the O/W emulsion. The amount of acyclovir in ester form that crossed the membrane into the receptor phase was measured at 24 h.

Conclusion

The synthesis of acyclovir esters—useful as prodrugs—have been performed by different methodologies more environmental friendly than those described in the literature. The substitution of dermotoxic and reprotoxic DMF by DMSO, a common permeation enhancer, as the reaction solvent let us to complete the synthesis in 10 min. The isolation of the monoesters as the unique product, was performed using water as a precipitating agent. The scale up of the synthesis of acyclovir hexanoate (**3c**) up to 300 g proves the feasabilty to implement it in industrial settings. The syntheses of medium and long chain esters of acyclovir (**3c**—**f**) carried out using DMAC10 solvents from renewable sources, rendered almost total conversion yields, yet at slower reaction rates but the selectivity towards a single monoacylated product was retained. All the acyclovir esters syntheses were also performed without any solvent, using the melted acid anhydrides as solvent medium. For acyclovir dodecanoate and tetradecanoate, a reduction in the substrates ratio up to 1/1 was succesfully achieved.

According to our results in the prodrug release assays, acyclovir hexanoate and dodecanoate achieved the highest transfer amount of total acyclovir into the receptor phase from our oil/water emulsion formulation. However, when an hydrogel formulation is used as a vehicle instead of the O/W emulsion, the transfer amount of total acyclovir reaches more than double in the case of ACVC6.

As solvent DMAC10 may be used as a permeation enhancer in transdermic applications, no need for complete solvent removal might be required from the reaction. This allowed us to obtain acyclovir prodrugs with a promotion enhancer included, in a mixture ready to be used in the following pharmaceutic preformulation stages. The permeation enhancement observed with the use of solvent DMAC10 (5% w/w) in the O/W emulsion formulation was around 10% compared to a blank emulsion. All the improvements stated in this work, which include the use of green solvents and reductions in the use of raw materials, solvents and energy, represent sensible advances for the synthesis of acyclovir prodrugs.

Experimental

Acyclovir (**1**) was acquired from Cinfa S.A. Acid anhydrides, *N*,*N*-dimethyl formamide and dimethyl sulfoxide were purchased from Sigma-Aldrich. 4-*N*,*N*-Dimethylamino pyridine purchased from VWR International Eurolab S.L. Solvents from renewable sources, *N*,*N*-dimethylamide of decanoic acid (DMAC10) and the mixture of *N*,*N*-dimethylamides from octanoic, decanoic and dodecanoic acid (DMAA) were supplied from Cognis Iberica S.L. Reaction progress was followed by TLC on Macherey Nagel Alugram Sil G/UV254 sheets visualized under UV light, by with HPLC on a Shimadzu CBM-20A using a Teknokroma C18 (30 cm, 4.6 mm, 5 μM) column. NMR spectra of samples on DMSO-d_6 for all products synthesized in DMSO were recorded in a Brucker Avance AC-250 (250 MHz) spectometer. Elemental analyses were performed by the C.A.I. of Micro-analysis Elemental de la Universidad Complutense de Madrid on a Leco CHNS 932. After verifying the identity of products synthesized in DMSO by NMR, HPLC and Elemental analysis, the following identification for products synthesized in renewable solvents and with no solvent, was only done by HPLC analysis.

Synthesis in DMSO of the Acyclovir Esters. Acyclovir Acetate (3a) 2-Amino-9-[(2-acetyloxy)ethyl-oxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The synthesis was carried out in a small flask under argon atmosphere. 2-Amino-1,9-dihydro-9-[(2-hydroxyyethoxy)methyl]-6*H*-purin-6-one, (**1**, acyclovir), (337 mg, 1.5 mmol) was dissolved in 6 ml of anhydrous DMSO and 4-*N*,*N*-dimethylamino-pyridine (DMAP) (18 mg, 0.15 mmol) was added. Argon atmosphere was set and the solution was stirred until reactants were dissolved. The acetic anhydride (0.42 ml, 4.5 mmol) was added keeping the inert atmosphere. After 10 min, stirring is stopped and the reaction volume is transferred to a 10 ml syringe. The reaction medium was added dropwise to a vigorously agitated beaker with 600 ml of water (100 reaction volumes). The supernatant precipitate was then filtered and dried to r.t. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.6—the numbers of carbons and nitrogen in H-NMR are in accordance with IUPAC acyclovir nomenclature—(1H, s broad, N1-H); 7.85 (1H s broad, C8H); 6.54 (2H, s broad, $\underline{NH_2}$); 5.36 (2H, s, C10$\underline{H_2}$); 4.07 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.66 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 1.96 (3H, t, ^3J=7 Hz, OOC-$\underline{CH_3}$). Elemental analysis Calcd for $C_{10}H_{13}N_5O_4$: C, 44.93; H, 4.90; N, 26.22. Found: C, 44.61, H: 5.01, N: 26.45.

Acyclovir Butyrate (3b) 2-Amino-9-[(2-butanoyloxy)ethyl-oxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The synthesis was carried out under the same conditions as in **3a**. The butyric anhydride (0.716 ml, 4.5 mmol) was added keeping the inert atmosphere. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.7 (1H, s broad, N1-H); 7.94 (1H, s, C8H); 6.58 (2H, s broad, $\underline{NH_2}$); 5.36 (2H, s, C10$\underline{H_2}$); 4.09 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.67 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 2.20 (2H, t, ^3J=7 Hz, OOC-$\underline{CH_2}$); 1.47 (2H, c, ^3J=7 Hz, OOC-CH$_2\underline{CH_2}$); 0.83 (3H, t, ^3J=7 Hz, OOC-(CH$_2$)$_2$-$\underline{CH_3}$). Elemental analysis Calcd for $C_{12}H_{17}N_5O_4$: C, 48.79; H, 5.80; N, 23.73. Found: C, 48.4, H, 5.89; N, 23.56.

Acyclovir Hexanoate (3c) 2-Amino-9-[(2-hexanoyloxy)ethyl-oxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The synthesis was carried out under the same conditions as **3a**. The hexanoic anhydride (1.03 ml, 4.5 mmol) was added keeping the inert atmosphere. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.7 (1H, s broad, N1-H); 7.84 (1H, s, C8H); 6.54 (2H, s broad, $\underline{NH_2}$); 5.35 (2H, s, C10$\underline{H_2}$); 4.09 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.66 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 2.22 (2H, t, ^3J=7 Hz, OOC-$\underline{CH_2}$); 1.47 (2H, q, ^3J=7 Hz, OOC-CH$_2\underline{CH_2}$); 1.22 (4H, mult, 2$\underline{CH_2}$); 0.84 (3H, t, ^3J=7 Hz, OOC-(CH$_2$)$_4\underline{CH_3}$). Elemental analysis Calcd for $C_{14}H_{21}N_5O_4$: C, 51.99; H 6.54; N, 21.67. Found: C, 51.76; H, 6.68; N, 21.44.

Acyclovir Decanoate (3d) 2-Amino-9-[(2-decanoyloxy)ethyl-oxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The reaction was carried out under the same conditions as **3a**. The decanoic anhydride (1.65 ml, 4.5 mmol) was added keeping the inert atmosphere. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.7 (1H, s broad, N1-H); 7.82 (1H, s, C8H); 6.60 (2H, s broad, $\underline{NH_2}$); 5.35 (2H, s, C10$\underline{H_2}$); 4.09 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.66 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 2.22 (2H, mult, OOC-$\underline{CH_2}$); 1.35 (2H, mult, OOC-CH$_2\underline{CH_2}$); 1.35—1.24 (12H, mult, 6$\underline{CH_2}$); 0.84 (3H, t, OOC(CH$_2$)$_8\underline{CH_3}$). Elemental analysis Calcd for $C_{18}H_{29}N_5O_4$: C, 59.96; H, 7.70; N, 18.47. Found: C, 59.69; H, 7.85; N, 18.29.

Acyclovir Dodecanoate (3e) 2-Amino-9-[(2-dodecanoyloxy)ethyl-oxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The synthesis was carried out under the same conditions as **3a**. The dodecanoic anhydride (1.72 g, 4.5 mmol) was added keeping the inert atmosphere. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.65 (1H, s broad, N1-H); 7.82 (1H, s, C8H); 6.53 (2H, s broad, $\underline{NH_2}$); 5.35 (2H, s, C10$\underline{H_2}$); 4.08 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.66 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 2.22 (2H, mult, OOC-$\underline{CH_2}$); 1.35 (2H, comp, OOC-CH$_2$CH$_2$); 1.45—1.24 (16H, mult, 8$\underline{CH_2}$); 0.83 (3H, t, OOC(CH$_2$)$_{10}\underline{CH_3}$). Elemental analysis Calcd for $C_{20}H_{33}N_5O_4$: C, 58.93; H, 8.16; N, 17.19. Found: C, 58.78; H, 8.28; N, 17.01.

Acyclovir Tetradecanate (3f) 2-Amino-9-[(2-tetradecanoyloxy)ethyloxy-methyl]-1,9-dihydro-6*H*-purin-6-one: The synthesis was carried out under the same conditions as **3a**. The dodecanoic anhydride (1.97 g, 4.5 mmol) was added keeping the inert atmosphere. ^{1}H-NMR (250 MHz, DMSO-d_6) δ: 10.7 (1H, s broad, N1-H); 7.82 (1H, s, C8H); 6.56 (2H, s broad, $\underline{NH_2}$); 5.35 (2H, s, C10H_2); 4.09 (2H, t, ^3J=7 Hz, C13$\underline{H_2}$); 3.66 (2H, t, ^3J=7 Hz, C12$\underline{H_2}$); 2.22 (2H, mult, OOC-$\underline{CH_2}$); 2.17 (2H, mult, OOC-$CH_2\underline{CH_2}$; 1.35 (20H, mult, 10$\underline{CH_2}$); 0.84 (3H, t, OOC$(CH_2)_{12}\underline{CH_3}$). Elemental analysis Calcd for $C_{22}H_{37}N_5O_4$: C, 60.65; H, 8.56; N, 16.09. Found: C, 60.38; H, 8.67; N, 15.89.

Synthesis in Solvents from Biomass. Acyclovir Hexanoate (3c) Acyclovir, 1.97 g (8.75 mmol); hexanoic anhydride, 3.71 g (17.5 mmol); DMAP, 0.108 g (0.88 mmol); DMAC10, DMAA or DMALA, 10 ml. Reaction temperature was set to 40 °C. Reactants were agitated during 24 h. In the purification process 30 volumes (300 ml) of a 50/50 acetonitrile/water mixture was added to the reaction mixture mainly to remove the DMAP and kept in agitation until a homogeneous suspension was observed. The suspension was filtered and allowed to dry to room temperature. HPLC method: isocratic water/acetonitrile 60/40, 0.8 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 6.2 min.

Acyclovir Decanoate (3d) Acyclovir, 1.97 g (8.75 mmol); decanoic anhydride, 5.71 g (17.5 mmol); DMAP, 0.108 g (0.88 mmol); DMAC10, DMAA or DMALA, 20 ml and the reaction temperature was set to 40 °C. Reactants were agitated during 24 h. In the purification process 30 volumes (300 ml) of a 50/50 acetonitrile/water mixture was added to the reaction mixture mainly to remove the DMAP and kept in agitation until a homogeneous suspension was observed. The suspension was filtered and allowed to dry to ambient temperature. HPLC method: isocratic water/acetonitrile 25/75, 1.0 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 4.6 min.

Acyclovir Dodecanoate (3e) Acyclovir, 1.97 g (8.75 mmol); dodecanoic anhydride, 6.50 g (17.5 mmol); DMAP, 0.108 g (0.88 mmol); DMAC10, DMAA or DMALA 20 ml. Reaction temperature was set to 40 °C. Reactants were agitated during 24 h. In the purification process 30 volumes (300 ml) of a 50/50 acetonitrile/water mixture was added to the reaction mixture mainly to remove the DMAP and kept in agitation until a homogeneous suspension was observed. The suspension was filtered and allowed to dry to ambient temperature. HPLC method: isocratic water/acetonitrile 25/75, 1.0 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 7.1 min.

Acyclovir Tetradecanoate (3f) Acyclovir, 1.97 g (8.75 mmol); decanoic anhydride, 7.67 g (17.5 mmol); DMAP, 0.108 g (0.88 mmol); DMAC10, DMAA or DMALA 20 ml. Reaction temperature was set to 40 °C. Reactants were agitated during 24 h. In the purification process 30 volumes (300 ml) of a 50/50 acetonitrile/water mixture was added to the reaction mixture mainly to remove the DMAP and kept in agitation until a homogeneous suspension was observed. The suspension was filtered and allowed to dry to ambient temperature. HPLC method: isocratic water/acetonitrile 10/90, 1.3 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 5.7 min.

Synthesis without Solvents. Acyclovir Dodecanoate (3e) Acyclovir, 1.97 g (8.75 mmol); dodecanoic anhydride, 3.25 g (17.5 mmol); DMAP, 0.108 g (0.88 mmol); reaction temperature was 30 °C. After 30 min the reaction was finished resulting in a 98% yield analyzed by HPLC. The purification process comprised the same steps as those used when DMSO was used as solvent. HPLC method: isocratic water/acetonitrile 25/75, 1.0 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 7.1 min.

Acyclovir Tetradecanoate (3f) Acyclovir, 1.97 g (8.75 mmol); decanoic anhydride, 3.83 g (8.75 mmol); DMAP, 0.108 g (0.88 mmol); reaction temperature was 30 °C. After 30 min of agitating reactants, the reaction was finished resulting in a 98% yield analyzed by HPLC. The purification stage comprised the same steps as those used when DMSO was used as solvent. HPLC method: isocratic water/acetonitrile 10/90, 1.3 ml/min on a C18 30 cm 4.6 mm 5 μm column. Retention time: 5.7 min.

Prodrug Release Assays The assays were performed in a Microettee Plus Hanson Research, (U.S.A.) at 32±1.5 °C using polysulfone synthetic membranes (Tuffryn®). Donor phases, either O/W23 emulsion or Hydrogel24 contained acyclovir ester at 5% w/w. Donor chamber was filled with 300 mg of donor phase. Receptor phase was phosphate buffer 0.15 M pH 7.4. Receptor phase samples were taken at 0.25, 0.5, 0.75, 1, 2, 4, 6, 8, 10, 12 and 24 h. For the prodrug release assay using solvent DMAC10, it was added to the O/W emulsion at 5% at the time of preparation.

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